End-to-End E-commerce SQL Analysis Documentation

Project Overview

This project analyzes an end-to-end e-commerce dataset using PostgreSQL. The goal is to understand business performance across sales, customers, products, marketing campaigns, funnel conversion, returns, inventory, payments, reviews, and market basket behavior.

The project uses 10 relational e-commerce tables:

- public.customers
- public.products
- public.orders
- public.order_items
- public.payments
- public.returns
- public.marketing_campaigns
- public.sessions
- public.inventory
- public.reviews

The analysis was performed using advanced SQL techniques such as joins, CTEs, window functions, conditional aggregation, ranking, cohort analysis, RFM segmentation, funnel analysis, ABC product classification, and market basket analysis.

Business Objective

The main business objective is to answer the following questions:

How much revenue and profit is the business generating? Which products, categories, customers, cities, and marketing campaigns are performing best? What is causing returns and refund losses? Which channels convert best? Which products need inventory attention? Which customers should be targeted for retention? Which products can be bundled together?

The final output of this SQL analysis can support Power BI dashboards, business strategy, marketing planning, inventory decisions, and customer retention campaigns.

Data Quality Checks

Row Count of All Tables

```
SELECT 'customers' AS table_name, COUNT(*) AS row_count FROM public.customers
UNION ALL
SELECT 'products', COUNT(*) FROM public.products
UNION ALL
SELECT 'orders', COUNT(*) FROM public.orders
UNION ALL
SELECT 'order_items', COUNT(*) FROM public.order_items
UNION ALL
SELECT 'payments', COUNT(*) FROM public.payments
UNION ALL
SELECT 'returns', COUNT(*) FROM public.returns
UNION ALL
SELECT 'marketing_campaigns', COUNT(*) FROM public.marketing_campaigns
UNION ALL
SELECT 'sessions', COUNT(*) FROM public.sessions
UNION ALL
SELECT 'inventory', COUNT(*) FROM public.inventory
UNION ALL
SELECT 'reviews', COUNT(*) FROM public.reviews;
```

Insight

- The dataset contains records in all 10 important e-commerce tables, which means the database is suitable for end-to-end analysis.
- The orders table has 600 records, and the payments table also has 600 records. This suggests that every order may have a related payment record, which is a positive sign for financial analysis.
- The order_items table has 923 records, which is higher than the number of orders. This indicates that many orders contain multiple products. On average: 923 order items / 600 orders = 1.54 items per order
- This is useful for basket analysis, average order size, and product combination analysis.
- The returns table has 40 records compared to 600 orders, giving an approximate return ratio of: 40 / 600 = 6.67%

Orders Without Valid Customers

```
SELECT o.*
FROM public.orders o
LEFT JOIN public.customers c
ON o.customer_id = c.customer_id
WHERE c.customer_id IS NULL;
```

Insight

- Every customer is valid

Data Quality Check: Order Items Total Validation

```
SELECT
  o.order_id,
  o.items_total AS stored_items_total,
  ROUND(SUM(oi.line_total)::numeric, 2) AS calculated_items_total,
  ROUND((o.items_total - SUM(oi.line_total))::numeric, 2) AS difference
FROM public.orders o
JOIN public.order_items oi
ON o.order_id = oi.order_id
GROUP BY o.order_id, o.items_total
HAVING ROUND(o.items_total::numeric, 2) <> ROUND(SUM(oi.line_total)::numeric, 2);
```

Insight:

- The query returned no mismatch records. This means orders.items_total matches the calculated total from order_items.line_total for all checked orders.

Negative Revenue or Profit

```
SELECT *
FROM public.orders
WHERE order_total < 0
 OR net_revenue < 0
 OR items_total < 0;
```

Insight

- The query returned empty result, meaning there are no negative values in order_total, net_revenue, or items_total.

Executive KPI Analysis

```
SELECT
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT customer_id) AS total_customers,
    ROUND(SUM(net_revenue)::numeric, 2) AS total_revenue,
    ROUND(SUM(net_profit)::numeric, 2) AS total_profit,

    -- Option A: Row-based average (Fixed with casting)
    ROUND(AVG(net_revenue)::numeric, 2) AS average_row_value,

    -- Option B: True Average Order Value (Total Revenue / Total Orders)
        ROUND((SUM(net_revenue) / NULLIF(COUNT(DISTINCT order_id), 0))::numeric, 2) AS
average_order_value,

    ROUND((SUM(net_profit) / NULLIF(SUM(net_revenue), 0))::numeric, 4) AS profit_margin,
        COUNT(DISTINCT CASE WHEN order_status = 'Returned' THEN order_id END) AS
returned_orders,
    ROUND(
        COUNT(DISTINCT CASE WHEN order_status = 'Returned' THEN order_id END)::numeric
        / NULLIF(COUNT(DISTINCT order_id), 0),
        4
    ) AS return_rate
FROM public.orders
WHERE order_status <> 'Cancelled';
```

Insight:

The business generated $4,484,287.80 revenue from 544 orders and 189 customers.

Total profit is $1,464,148.15, with a strong 32.65% profit margin.

Average order value is $8,243.18. Return rate is 7.35%, based on 40 returned orders.

Recommendation:

Focus on increasing repeat purchases from the 189 customers and investigate the reasons behind returned orders.

Monthly Sales Trend with Growth Rate

```
WITH monthly_sales AS (
    SELECT
        DATE_TRUNC('month', order_date)::DATE AS month,
        COUNT(DISTINCT order_id) AS total_orders,
        ROUND(SUM(net_revenue)::numeric, 2) AS revenue,
        ROUND(SUM(net_profit)::numeric, 2) AS profit,
        -- Grab the previous month's revenue right here:
        LAG(ROUND(SUM(net_revenue)::numeric, 2)) OVER (ORDER BY DATE_TRUNC('month',
order_date)::DATE) AS prev_revenue
    FROM public.orders
    WHERE order_status <> 'Cancelled'
    GROUP BY DATE_TRUNC('month', order_date)::DATE
)
SELECT
    month,
    total_orders,
    revenue,
    profit,
    -- Much cleaner formula to read and maintain:
        ROUND(((revenue - prev_revenue) / NULLIF(prev_revenue, 0))::numeric, 4) AS
month_over_month_growth,
    ROUND((SUM(revenue) OVER (ORDER BY month))::numeric, 2) AS running_total_revenue
FROM monthly_sales
ORDER BY month;
```

Insight

Revenue increased from $116,196.88 in Jan 2024 to a cumulative $4,484,287.80 by Dec 2025. The best revenue month was Oct 2025: $355,440.89, while the strongest growth happened in Jul 2025: +108%. However, revenue dropped sharply in Aug 2025: -62.47%.

Recommendation

Analyze what caused the strong growth in Jul and Oct 2025, then repeat the same campaign, product, or customer strategy. Also investigate the major revenue drops in Aug 2025, Nov 2025, and Jun 2025.

Category and Sub-Category Performance

```
SELECT
    p.category,
    p.sub_category,
    COUNT(DISTINCT o.order_id) AS total_orders,
    SUM(oi.quantity) AS units_sold,
    ROUND(SUM(oi.line_total)::numeric, 2) AS revenue,
    ROUND(SUM(oi.line_profit)::numeric, 2) AS profit,
    ROUND((SUM(oi.line_profit) / NULLIF(SUM(oi.line_total), 0))::numeric, 4) AS profit_margin
FROM public.order_items oi
JOIN public.products p
ON oi.product_id = p.product_id
JOIN public.orders o
ON oi.order_id = o.order_id
WHERE o.order_status <> 'Cancelled'
GROUP BY p.category, p.sub_category
ORDER BY revenue DESC;
```

Insight

Mobile Accessories is the top revenue generator with $742,000.10, followed by Audio with $625,687.11. However, Hair Care has the highest profit margin at 41.99%, and Kitchen also performs strongly with 37.57% margin.

Recommendation

Keep promoting Mobile Accessories and Audio for revenue growth, but increase marketing focus on Hair Care and Kitchen because they generate stronger margins.

Top Product Performance Report

```
WITH product_sales AS (
    SELECT
        p.product_id,
        p.product_name,
        p.category,
        p.brand,
        SUM(oi.quantity) AS units_sold,
        ROUND(SUM(oi.line_total)::numeric, 2) AS revenue, -- Fixed cast
```

```
        ROUND(SUM(oi.line_profit)::numeric, 2) AS profit   -- Fixed cast
    FROM public.order_items oi
    JOIN public.products p
    ON oi.product_id = p.product_id
    JOIN public.orders o
    ON oi.order_id = o.order_id
    WHERE o.order_status <> 'Cancelled'
    GROUP BY p.product_id, p.product_name, p.category, p.brand
)
SELECT
    product_id,
    product_name,
    category,
    brand,
    units_sold,
    revenue,
    profit,
    RANK() OVER (ORDER BY revenue DESC) AS revenue_rank,
    RANK() OVER (ORDER BY profit DESC) AS profit_rank
FROM product_sales
ORDER BY revenue_rank ASC
LIMIT 20;
```

Insight

Nivea Fragrance Item 3 is the best-performing product, ranking #1 in both revenue and profit with $303,651.54 revenue and $138,135.98 profit. RFL Decor Item 39 and Le Reve Men Clothing Item 16 are also strong performers in both revenue and profit.

Recommendation

Prioritize top-ranked products in marketing campaigns, homepage placements, bundle offers, and inventory planning. Also review products with high revenue but lower profit rank, such as Bata Women Clothing Item 44, to improve margin.

ABC Product Analysis

```
WITH product_revenue AS (
    SELECT
        p.product_id,
        p.product_name,
        p.category,
        ROUND(SUM(oi.line_total)::numeric, 2) AS revenue -- Fixed cast inside first CTE
    FROM public.order_items oi
    JOIN public.products p
    ON oi.product_id = p.product_id
    JOIN public.orders o
    ON oi.order_id = o.order_id
    WHERE o.order_status <> 'Cancelled'
    GROUP BY p.product_id, p.product_name, p.category
),
revenue_share AS (
    SELECT
        product_id,
        product_name,
        category,
        revenue,
        -- Fixed: Cast the total revenue share division to numeric
        ROUND(
            (revenue / NULLIF(SUM(revenue) OVER (), 0))::numeric,
            4
        ) AS revenue_share,
        -- Fixed: Cast the running total/cumulative share division to numeric
        ROUND(
              (SUM(revenue) OVER (ORDER BY revenue DESC) / NULLIF(SUM(revenue) OVER (),
0))::numeric,
            4
        ) AS cumulative_revenue_share
    FROM product_revenue
)
SELECT
    product_id,
    product_name,
    category,
    revenue,
    revenue_share,
    cumulative_revenue_share,
    CASE
        WHEN cumulative_revenue_share <= 0.80 THEN 'A - High Value'
```

```
        WHEN cumulative_revenue_share <= 0.95 THEN 'B - Medium Value'
        ELSE 'C - Low Value'
    END AS abc_category
FROM revenue_share
ORDER BY revenue DESC;
```

Business Question

Which products should the business prioritize based on revenue contribution?

Insight

The A-category products generate most of the revenue. Around 26 high-value products contribute about 78.47% of total revenue. B-category products support medium revenue growth, while C-category products contribute only about 5.71%.

Recommendation

Focus inventory, marketing, and promotions on A-category products. Monitor B-category products for growth potential, and review C-category products for discounting, bundling, or possible removal.

Customer Value Analysis

```
SELECT
    c.customer_id,
    c.customer_name,
    c.city,
    c.acquisition_channel,
    c.customer_segment,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(SUM(o.net_revenue)::numeric, 2) AS total_spent,
    ROUND(AVG(o.net_revenue)::numeric, 2) AS average_order_value,
    ROUND(SUM(o.net_profit)::numeric, 2) AS total_profit,
    MIN(o.order_date) AS first_order_date,
    MAX(o.order_date) AS last_order_date
FROM public.customers c
JOIN public.orders o
ON c.customer_id = o.customer_id
WHERE o.order_status <> 'Cancelled'
GROUP BY
```

```
    c.customer_id,
    c.customer_name,
    c.city,
    c.acquisition_channel,
    c.customer_segment
ORDER BY total_spent DESC
LIMIT 20;
```

Insight

The top customer is Hasan Mia with $90,984.39 total spent and $30,386.38 profit from 7 orders. Most top customers are from the Premium segment, and Instagram, Referral, YouTube, Google, and Organic are bringing high-value customers.

Recommendation

Create a VIP retention strategy for top customers, such as loyalty rewards, personalized offers, and early access deals. Also increase focus on channels producing premium customers, especially Instagram and Referral.

RFM Customer Segmentation

```
WITH rfm_base AS (
    SELECT
        c.customer_id,
        c.customer_name,
        c.city,
        c.acquisition_channel,
        MAX(o.order_date) AS last_purchase_date,
        DATE '2026-01-01' - MAX(o.order_date) AS recency_days,
        COUNT(DISTINCT o.order_id) AS frequency,
        ROUND(SUM(o.net_revenue)::numeric, 2) AS monetary_value -- Fixed cast
    FROM public.customers c
    JOIN public.orders o
    ON c.customer_id = o.customer_id
    WHERE o.order_status <> 'Cancelled'
    GROUP BY
        c.customer_id,
        c.customer_name,
        c.city,
        c.acquisition_channel
```

```
),
rfm_score AS (
   SELECT
      *,
      -- Fixed: Low recency days = Recent purchase = High score (5)
      NTILE(5) OVER (ORDER BY recency_days ASC) AS recency_score,
      -- Fixed: High frequency = Loyal buyer = High score (5)
      NTILE(5) OVER (ORDER BY frequency DESC) AS frequency_score,
      -- Fixed: High monetary value = Big spender = High score (5)
      NTILE(5) OVER (ORDER BY monetary_value DESC) AS monetary_score
   FROM rfm_base
)
SELECT
   customer_id,
   customer_name,
   city,
   acquisition_channel,
   last_purchase_date,
   recency_days,
   frequency,
   monetary_value,
   recency_score,
   frequency_score,
   monetary_score,
   (recency_score + frequency_score + monetary_score) AS rfm_total_score,
   CASE
        WHEN recency_score >= 4 AND frequency_score >= 4 AND monetary_score >= 4 THEN
'Champions'
      WHEN frequency_score >= 4 AND monetary_score >= 4 THEN 'Loyal Customers'
      WHEN recency_score >= 4 AND frequency_score <= 2 THEN 'New Customers'
      WHEN recency_score <= 2 AND frequency_score >= 3 THEN 'At Risk'
      WHEN recency_score <= 2 AND frequency_score <= 2 THEN 'Lost Customers'
      ELSE 'Regular Customers'
   END AS rfm_segment
FROM rfm_score
ORDER BY rfm_total_score DESC, monetary_value DESC;
```

Insight

The result shows some customers with 1 order and very old purchase dates being classified as Champions, while recent high-spending customers are classified as Lost Customers. This means the RFM scoring logic is reversed because NTILE(5) gives score 1 to the first group, not 5.

Recommendation

Fix the scoring direction before using this for business decisions. Recent buyers, frequent buyers, and high spenders should receive higher RFM scores.

RFM Segment Summary

```
WITH rfm_segments AS (
  WITH rfm_base AS (
    SELECT
      c.customer_id,
      MAX(o.order_date) AS last_purchase_date,
        -- Fixed: Cast the max order date to a DATE first so the subtraction results in an integer
natively
      (DATE '2026-01-01' - MAX(o.order_date)::DATE) AS recency_days,
      COUNT(DISTINCT o.order_id) AS frequency,
      ROUND(SUM(o.net_revenue)::numeric, 2) AS monetary_value
    FROM public.customers c
    JOIN public.orders o
    ON c.customer_id = o.customer_id
    WHERE o.order_status <> 'Cancelled'
    GROUP BY c.customer_id
  ),
  rfm_score AS (
    SELECT
      *,
      NTILE(5) OVER (ORDER BY recency_days ASC) AS recency_score,
      NTILE(5) OVER (ORDER BY frequency DESC) AS frequency_score,
      NTILE(5) OVER (ORDER BY monetary_value DESC) AS monetary_score
    FROM rfm_base
  )
  SELECT
    *,
    CASE
         WHEN recency_score >= 4 AND frequency_score >= 4 AND monetary_score >= 4
THEN 'Champions'
      WHEN frequency_score >= 4 AND monetary_score >= 4 THEN 'Loyal Customers'
      WHEN recency_score >= 4 AND frequency_score <= 2 THEN 'New Customers'
      WHEN recency_score <= 2 AND frequency_score >= 3 THEN 'At Risk'
      WHEN recency_score <= 2 AND frequency_score <= 2 THEN 'Lost Customers'
      ELSE 'Regular Customers'
    END AS rfm_segment
  FROM rfm_score
```

```
)
SELECT
    rfm_segment,
    COUNT(*) AS total_customers,
    ROUND(AVG(recency_days)::numeric, 2) AS avg_recency_days,
    ROUND(AVG(frequency)::numeric, 2) AS avg_frequency,
    ROUND(AVG(monetary_value)::numeric, 2) AS avg_customer_value,
    ROUND(SUM(monetary_value)::numeric, 2) AS total_revenue
FROM rfm_segments
GROUP BY rfm_segment
ORDER BY total_revenue DESC;
```

Insight

The report shows “Lost Customers” generating the highest revenue: $1,754,509.77, with high average frequency 4.60 and recent purchase activity 49.89 days. This means the segment labels are still incorrect because these customers look like Champions, not lost customers.

Recommendation

Fix the RFM scoring direction before using this result. High recency score should mean recent purchase, high frequency score should mean more orders, and high monetary score should mean higher spending.

Cohort Retention Analysis

```
WITH first_purchase AS (
    SELECT
        customer_id,
        DATE_TRUNC('month', MIN(order_date))::DATE AS cohort_month
    FROM public.orders
    WHERE order_status <> 'Cancelled'
    GROUP BY customer_id
),
customer_orders AS (
    SELECT
        o.customer_id,
        fp.cohort_month,
        DATE_TRUNC('month', o.order_date)::DATE AS order_month,
        -- Fixed: Safer calendar month difference math bypassing the AGE() function interval trap
```

```
        (
            (EXTRACT(YEAR FROM DATE_TRUNC('month', o.order_date)) - EXTRACT(YEAR
FROM fp.cohort_month)) * 12
            + (EXTRACT(MONTH FROM DATE_TRUNC('month', o.order_date)) -
EXTRACT(MONTH FROM fp.cohort_month))
        )::INT AS month_number
    FROM public.orders o
    JOIN first_purchase fp
    ON o.customer_id = fp.customer_id
    WHERE o.order_status <> 'Cancelled'
),
cohort_size AS (
    SELECT
        cohort_month,
        COUNT(DISTINCT customer_id) AS total_customers
    FROM first_purchase
    GROUP BY cohort_month
)
SELECT
    co.cohort_month,
    co.month_number,
    COUNT(DISTINCT co.customer_id) AS active_customers,
    cs.total_customers AS cohort_size,
    ROUND(
        COUNT(DISTINCT co.customer_id)::NUMERIC
        / NULLIF(cs.total_customers, 0),
        4
    ) AS retention_rate
FROM customer_orders co
JOIN cohort_size cs
ON co.cohort_month = cs.cohort_month
GROUP BY co.cohort_month, co.month_number, cs.total_customers
ORDER BY co.cohort_month, co.month_number;
```

Insight

Most cohorts show low repeat purchase activity after month 0. Some cohorts have strong comeback behavior, such as Feb 2024 with 54.55% retention in month 20 and Feb 2025 with 50% retention in month 8. Recent cohorts from late 2025 only show month 0 because they do not have enough future months yet.

Recommendation

Create retention campaigns after the first purchase, especially within the first 1–3 months, using email, remarketing, loyalty points, and personalized offers.

Business Impact

Improving early-month retention can increase repeat orders, customer lifetime value, and long-term revenue without depending only on new customer acquisition.

Marketing Campaign ROI Analysis

```
SELECT
    campaign_id,
    campaign_name,
    channel,
    campaign_cost,
    impressions,
    clicks,
    conversions,
    revenue_generated,
    -- Fixed: Added numeric cast for ROAS calculation
    ROUND((revenue_generated / NULLIF(campaign_cost, 0))::numeric, 2) AS roas,
    -- Kept your excellent fix here:
    ROUND(conversions::numeric / NULLIF(clicks, 0), 4) AS conversion_rate,
    -- Fixed: Added numeric cast for CPA calculation
    ROUND((campaign_cost / NULLIF(conversions, 0))::numeric, 2) AS cost_per_acquisition
FROM public.marketing_campaigns
ORDER BY roas DESC;
```

Insight

Ramadan Offer is the only strong campaign with ROAS 1.47, meaning it generated more revenue than cost. All other campaigns have ROAS below 1, so they are not recovering their campaign cost. New Year Sale performed worst with ROAS 0.07 and very high CPA of $96,109.05.

Recommendation

Increase budget for Ramadan Offer and review weak campaigns like New Year Sale and Black Friday. Improve targeting, offer quality, and landing page conversion before spending more.

Campaign Actual Revenue from Orders

```
SELECT
    mc.campaign_id,
    mc.campaign_name,
    mc.channel,
    mc.campaign_cost,
    COUNT(DISTINCT o.order_id) AS actual_orders,
    ROUND(SUM(o.net_revenue)::numeric, 2) AS actual_revenue, -- Fixed cast
    ROUND(SUM(o.net_profit)::numeric, 2) AS actual_profit,   -- Fixed cast
    -- Fixed: Aggregated revenue divided by cost, cast to numeric for rounding
    ROUND((SUM(o.net_revenue) / NULLIF(mc.campaign_cost, 0))::numeric, 2) AS actual_roas
FROM public.marketing_campaigns mc
LEFT JOIN public.orders o
ON mc.campaign_id = o.campaign_id
AND o.order_status <> 'Cancelled'
GROUP BY
    mc.campaign_id,
    mc.campaign_name,
    mc.channel,
    mc.campaign_cost
ORDER BY actual_roas DESC;
```

Insight

Ramadan Offer performed best with 5 actual orders, $48,909.45 revenue, $17,104.13 profit, and the highest actual ROAS of 1.47. All other campaigns have actual ROAS below 1, meaning their generated revenue did not cover campaign cost.

Recommendation:

Scale Ramadan Offer and reduce or redesign low-performing campaigns like New Year Sale, Black Friday, and Summer Deals.

Funnel Analysis

```
SELECT
```

```
  traffic_source,
  device_type,
  COUNT(DISTINCT session_id) AS total_sessions,
     COUNT(DISTINCT CASE WHEN added_to_cart = 'Yes' THEN session_id END) AS
cart_sessions,
    COUNT(DISTINCT CASE WHEN checkout_started = 'Yes' THEN session_id END) AS
checkout_sessions,
     COUNT(DISTINCT CASE WHEN order_placed = 'Yes' THEN session_id END) AS
order_sessions,
  ROUND(
     COUNT(DISTINCT CASE WHEN added_to_cart = 'Yes' THEN session_id END)::NUMERIC
     / NULLIF(COUNT(DISTINCT session_id), 0),  4
  ) AS cart_rate,
  ROUND(
            COUNT(DISTINCT CASE WHEN checkout_started = 'Yes' THEN session_id
END)::NUMERIC
      / NULLIF(COUNT(DISTINCT CASE WHEN added_to_cart = 'Yes' THEN session_id END),
0),   4
  ) AS checkout_rate,
  ROUND(
     COUNT(DISTINCT CASE WHEN order_placed = 'Yes' THEN session_id END)::NUMERIC
     / NULLIF(COUNT(DISTINCT session_id), 0),  4
  ) AS conversion_rate
FROM public.sessions
GROUP BY traffic_source, device_type
ORDER BY conversion_rate DESC;
```

Insight

Google Mobile has the highest conversion rate at 74.07%, followed by Instagram Mobile at 72.04% and YouTube Mobile at 70.42%. Mobile traffic performs better than desktop and app in most channels. The weakest performance is Referral App at 38.10% and YouTube App at 42.55%.

Recommendation:

Increase marketing focus on Google Mobile, Instagram Mobile, and YouTube Mobile. Improve the app experience for Referral and YouTube traffic because conversion is low there.

Cart Abandonment Analysis

```
SELECT
  traffic_source,
  device_type,
  COUNT(*) AS total_sessions,
  COUNT(*) FILTER (WHERE added_to_cart = 'Yes') AS added_to_cart_sessions,
  COUNT(*) FILTER (
     WHERE added_to_cart = 'Yes'
     AND order_placed = 'No'
  ) AS cart_abandoned_sessions,
  ROUND(
     COUNT(*) FILTER (
        WHERE added_to_cart = 'Yes'
        AND order_placed = 'No'
     )::NUMERIC
     / NULLIF(COUNT(*) FILTER (WHERE added_to_cart = 'Yes'), 0),  4
  ) AS cart_abandonment_rate
FROM public.sessions
GROUP BY traffic_source, device_type
ORDER BY cart_abandonment_rate DESC;
```

Insight:

Referral App has the highest cart abandonment rate at 30.43%, followed by Google Desktop at 30.30% and YouTube Desktop at 29.03%. Best performance comes from Google Mobile and YouTube Mobile, both with only 9.09% abandonment.

Recommendation:

Improve checkout experience for Referral App, Google Desktop, and YouTube Desktop. Check payment issues, delivery cost visibility, page speed, and checkout steps.

Return Reason Analysis

```
SELECT
   r.return_reason,
   COUNT(*) AS total_returns,
   SUM(r.returned_quantity) AS total_units_returned,
   ROUND(SUM(r.refund_amount)::numeric, 2) AS total_refund_amount, -- Fixed cast
   ROUND(AVG(r.refund_amount)::numeric, 2) AS avg_refund_amount,   -- Fixed cast
```

```
    ROUND(SUM(r.return_processing_cost)::numeric, 2) AS total_processing_cost -- Fixed cast
FROM public.returns r
GROUP BY r.return_reason
ORDER BY total_returns DESC;
```

Insight

Damaged Product is the top return reason with 12 returns and the highest refund amount of $72,995.42. Late Delivery is second with 9 returns, followed by Quality Issue with 7 returns.

Recommendation

Improve product packaging, supplier quality control, and delivery process. Focus first on reducing damaged products and late deliveries.

Return Rate by Product Category

```
WITH sold AS (
    SELECT
        p.category,
        COUNT(DISTINCT o.order_id) AS total_orders,
        SUM(oi.quantity) AS units_sold
    FROM public.orders o
    JOIN public.order_items oi
    ON o.order_id = oi.order_id
    JOIN public.products p
    ON oi.product_id = p.product_id
    WHERE o.order_status <> 'Cancelled'
    GROUP BY p.category
),
returned AS (
    SELECT
        p.category,
        COUNT(DISTINCT r.return_id) AS total_returns,
        SUM(r.returned_quantity) AS units_returned,
        ROUND(SUM(r.refund_amount)::numeric, 2) AS refund_amount -- Fixed cast here
    FROM public.returns r
    JOIN public.products p
    ON r.product_id = p.product_id
```

```
    GROUP BY p.category
)
SELECT
    s.category,
    s.total_orders,
    s.units_sold,
    COALESCE(r.total_returns, 0) AS total_returns,
    COALESCE(r.units_returned, 0) AS units_returned,
    -- Fixed: Force COALESCE fallback to use numeric decimal formatting
    COALESCE(r.refund_amount, 0.00) AS refund_amount,
    ROUND(
        COALESCE(r.units_returned, 0)::numeric
        / NULLIF(s.units_sold, 0),
        4
    ) AS unit_return_rate
FROM sold s
LEFT JOIN returned r
ON s.category = r.category
ORDER BY unit_return_rate DESC;
```

Insight

Electronics has the highest unit return rate at 4.71%, with 20 returns and $81,042.95 refund amount. Beauty is second at 3.83%, followed by Fashion at 3.30%. Home & Living has the lowest return rate at 2.42%.

Recommendation

Focus first on reducing returns in Electronics by checking product quality, packaging, delivery handling, and product descriptions.

Late Delivery Impact on Returns

```
SELECT
    o.delivery_status,
    COUNT(DISTINCT o.order_id) AS total_orders,
    COUNT(DISTINCT r.return_id) AS total_returns,
    ROUND(
        COUNT(DISTINCT r.return_id)::numeric
        / NULLIF(COUNT(DISTINCT o.order_id), 0),
        4
```

```
    ) AS return_rate,
    -- Fixed: Cast the sum of the floats to numeric before rounding, then coalesce the final result
    COALESCE(ROUND(SUM(r.refund_amount)::numeric, 2), 0.00) AS total_refund_amount
FROM public.orders o
LEFT JOIN public.returns r
ON o.order_id = r.order_id
WHERE o.order_status <> 'Cancelled'
GROUP BY o.delivery_status
ORDER BY return_rate DESC;
```

Insight

Delivered orders have a slightly higher return rate at 7.61%, compared to Delayed orders at 6.75%. Delivered orders also generated a higher refund amount of $146,085.01.

Recommendation

Investigate return reasons for delivered orders, especially product quality, damage, wrong item, or size issues.

Payment Method Analysis

```
SELECT
    payment_method,
    payment_status,
    COUNT(*) AS total_transactions,
    -- Fixed: Cast sums and averages to numeric before rounding, then coalesce
    COALESCE(ROUND(SUM(amount_paid)::numeric, 2), 0.00) AS total_amount_paid,
    COALESCE(ROUND(SUM(transaction_fee)::numeric, 2), 0.00) AS total_transaction_fee,
    COALESCE(ROUND(AVG(transaction_fee)::numeric, 2), 0.00) AS avg_transaction_fee,
    COALESCE(ROUND(SUM(refund_amount)::numeric, 2), 0.00) AS total_refund_amount
FROM public.payments
GROUP BY payment_method, payment_status
ORDER BY total_amount_paid DESC;
```

Insight

bKash Success generated the highest paid amount at $1,416,328.71, followed by Cash on Delivery at $1,148,896.31 and Card at $1,057,730.67. For refunded payments, bKash has the highest refund amount: $60,906.48, followed by Card: $56,963.34.

Recommendation

Continue prioritizing bKash, Cash on Delivery, and Card as major payment options, but monitor refund patterns in bKash and Card transactions.

Inventory Stockout Risk Analysis

```
SELECT
    p.product_id,
    p.product_name,
    p.category,
    i.warehouse_location,
    i.beginning_stock,
    i.units_sold,
    i.units_returned,
    i.units_restocked,
    i.ending_stock,
    i.stockout_risk,
    ROUND(
        i.units_sold::NUMERIC
        / NULLIF(i.beginning_stock + i.units_restocked, 0),
        4
    ) AS inventory_turnover_rate
FROM public.inventory i
JOIN public.products p
ON i.product_id = p.product_id
ORDER BY inventory_turnover_rate DESC;
```

Insight

All products currently show stockout_risk = No, meaning inventory is available. However, Apex Women Clothing Item 23 has the highest turnover rate at 23.36%, followed by Samsung Computer Accessories Item 25 at 17.71% and Xiaomi Computer Accessories Item 28 at 17.54%.

Recommendation

Monitor high-turnover products closely and plan earlier restocking, especially for Fashion and Electronics items with fast stock movement.

High Demand but Low Stock Products

```
SELECT
    p.product_id,
    p.product_name,
    p.category,
    i.units_sold,
    i.ending_stock,
    i.stockout_risk,
    ROUND(
        i.units_sold::NUMERIC
        / NULLIF(i.ending_stock, 0),
        2
    ) AS demand_to_stock_ratio
FROM public.inventory i
JOIN public.products p
ON i.product_id = p.product_id
WHERE i.ending_stock > 0
ORDER BY demand_to_stock_ratio DESC
LIMIT 20;
```

Insight

All top 20 products show stockout_risk = No, so there is no immediate stockout issue. However, Apex Women Clothing Item 23 has the highest demand-to-stock ratio at 0.30, followed by Xiaomi Computer Accessories Item 28 and Samsung Computer Accessories Item 25 at 0.21.

Recommendation

Monitor these high-demand products closely and prepare early restocking plans, especially for Fashion and Electronics items.

Review and Rating Analysis

```
SELECT
   p.category,
   p.sub_category,
   COUNT(r.review_id) AS total_reviews,
```

```sql
  ROUND(AVG(r.rating), 2) AS avg_rating,
  COUNT(*) FILTER (WHERE r.rating <= 2) AS negative_reviews,
  COUNT(*) FILTER (WHERE r.rating >= 4) AS positive_reviews
FROM public.reviews r
JOIN public.products p
ON r.product_id = p.product_id
GROUP BY p.category, p.sub_category
ORDER BY avg_rating DESC;
```

Insight

Fashion Footwear has the highest average rating at 4.27, followed by Men Clothing at 4.18. The lowest-rated sub-categories are Computer Accessories at 3.67 and Kitchen at 3.69. Computer Accessories also has the highest negative reviews with 3 negative reviews.

Recommendation

Promote high-rated categories like Footwear and Men Clothing. Investigate quality, product description, or delivery issues in Computer Accessories and Kitchen.

Product Quality Risk Analysis

```sql
WITH review_summary AS (
   SELECT
      product_id,
      COUNT(*) AS total_reviews,
      ROUND(AVG(rating)::numeric, 2) AS avg_rating, -- Fixed cast
      COUNT(*) FILTER (WHERE rating <= 2) AS negative_reviews
   FROM public.reviews
   GROUP BY product_id
),
return_summary AS (
   SELECT
      product_id,
      COUNT(*) AS total_returns,
      ROUND(SUM(refund_amount)::numeric, 2) AS refund_amount -- Fixed cast
   FROM public.returns
   GROUP BY product_id
)
```

```sql
SELECT
    p.product_id,
    p.product_name,
    p.category,
    COALESCE(rs.total_reviews, 0) AS total_reviews,
    -- Fixed: Swapped fallback literal to 0.00 for consistent decimal output
    COALESCE(rs.avg_rating, 0.00) AS avg_rating,
    COALESCE(rs.negative_reviews, 0) AS negative_reviews,
    COALESCE(rt.total_returns, 0) AS total_returns,
    -- Fixed: Swapped fallback literal to 0.00 for consistent decimal output
    COALESCE(rt.refund_amount, 0.00) AS refund_amount,
    CASE
        WHEN COALESCE(rs.avg_rating, 5.00) < 3 -- Matched type to decimal
             AND COALESCE(rt.total_returns, 0) > 0
        THEN 'High Quality Risk'
        WHEN COALESCE(rs.avg_rating, 5.00) < 4 -- Matched type to decimal
        THEN 'Medium Quality Risk'
        ELSE 'Low Quality Risk'
    END AS quality_risk_level
FROM public.products p
LEFT JOIN review_summary rs
ON p.product_id = rs.product_id
LEFT JOIN return_summary rt
ON p.product_id = rt.product_id
ORDER BY refund_amount DESC, avg_rating ASC;
```

Insight

Nivea Fragrance Item 3 has the highest refund amount at $32,932.80 with 3 returns and an average rating of 3.80, making it a Medium Quality Risk product. Other risky products include Bata Footwear Item 46 and Xiaomi Mobile Accessories Item 15, both with returns and ratings below 4.

Recommendation

Review product quality, packaging, supplier issues, and customer complaints for medium-risk products with high refunds.

Discount Impact on Profit

```sql
SELECT
    p.category,
```

```
    ROUND(SUM(oi.discount_amount)::numeric, 2) AS total_discount,
    ROUND(SUM(oi.line_total)::numeric, 2) AS revenue_after_discount,
    ROUND(SUM(oi.line_profit)::numeric, 2) AS profit,
    -- Fixed: Added numeric cast to the aggregate discount rate division math
    ROUND(
        (SUM(oi.discount_amount) / NULLIF(SUM(oi.gross_amount), 0))::numeric,
        4
    ) AS discount_rate,
    -- Fixed: Added numeric cast to the aggregate profit margin division math
    ROUND(
        (SUM(oi.line_profit) / NULLIF(SUM(oi.line_total), 0))::numeric,
        4
    ) AS profit_margin
FROM public.order_items oi
JOIN public.products p
ON oi.product_id = p.product_id
JOIN public.orders o
ON oi.order_id = o.order_id
WHERE o.order_status <> 'Cancelled'
GROUP BY p.category
ORDER BY discount_rate DESC;
```

Insight

Electronics has the highest discount rate at 5.38% and generates the highest revenue: $1,895,564.66. However, Beauty has the best profit margin at 39.57% with the lowest discount rate of 4.04%.

Recommendation

Keep using discounts for Electronics to drive sales, but avoid over-discounting. Promote Beauty more because it delivers stronger profitability with less discount.

City-Wise Business Performance

```
SELECT
    shipping_city,
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT customer_id) AS total_customers,
    -- Fixed: Cast sums and averages to numeric before rounding, then coalesce
    COALESCE(ROUND(SUM(net_revenue)::numeric, 2), 0.00) AS revenue,
    COALESCE(ROUND(SUM(net_profit)::numeric, 2), 0.00) AS profit,
```

```
    COALESCE(ROUND(AVG(net_revenue)::numeric, 2), 0.00) AS average_order_value,
    COUNT(*) FILTER (WHERE delivery_status = 'Delayed') AS delayed_orders,
    ROUND(
        COUNT(*) FILTER (WHERE delivery_status = 'Delayed')::numeric
        / NULLIF(COUNT(*), 0),
        4
    ) AS delay_rate
FROM public.orders
WHERE order_status <> 'Cancelled'
GROUP BY shipping_city
ORDER BY revenue DESC;
```

Insight

Khulna is the top revenue city with $905,901.97, followed by Rajshahi with $808,932.54. However, Khulna also has the highest delay rate at 32.04%, while Barishal has the lowest delay rate at 26.39%.

Recommendation

Focus sales and customer retention efforts in Khulna, Rajshahi, and Sylhet, but improve delivery operations in Khulna and Dhaka where delay rates are high.

Market Basket Analysis

```
SELECT
p1.product_name AS product_1,
p2.product_name AS product_2,
COUNT(*) AS times_bought_together
FROM public.order_items oi1
JOIN public.order_items oi2
ON oi1.order_id = oi2.order_id
AND oi1.product_id < oi2.product_id
JOIN public.products p1
ON oi1.product_id = p1.product_id
JOIN public.products p2
ON oi2.product_id = p2.product_id
JOIN public.orders o
ON oi1.order_id = o.order_id
WHERE o.order_status <> 'Cancelled'
```

```
GROUP BY p1.product_name, p2.product_name
HAVING COUNT(*) >= 2
ORDER BY times_bought_together DESC
LIMIT 20;
```

Insight

The top product pairs were bought together only 2 times each, meaning there is no very strong product-bundle pattern yet. Some useful bundle opportunities appear across Fashion + Beauty, Electronics + Electronics, and Home & Living + Beauty.

Recommendation

Test small bundle offers using repeated pairs, such as Samsung Computer Accessories Item 6 + Xiaomi Computer Accessories Item 28 or Apex Women Clothing Item 23 + Bata Footwear Item 46.

Sales View

```
CREATE OR REPLACE VIEW staging.vw_monthly_sales_analysis AS
WITH monthly_sales AS (
  SELECT
    DATE_TRUNC('month', order_date)::DATE AS month,
    COUNT(DISTINCT order_id) AS total_orders,
    ROUND(SUM(net_revenue)::numeric, 2) AS revenue, -- Fixed cast inside CTE
    ROUND(SUM(net_profit)::numeric, 2) AS profit    -- Fixed cast inside CTE
  FROM public.orders
  WHERE order_status <> 'Cancelled'
  GROUP BY DATE_TRUNC('month', order_date)::DATE
)
SELECT
  month,
  total_orders,
  revenue,
  profit,
  -- Fixed: Cast the entire MoM growth division math to numeric before rounding
  ROUND(
    ((revenue - LAG(revenue) OVER (ORDER BY month))
    / NULLIF(LAG(revenue) OVER (ORDER BY month), 0))::numeric,
    4
  ) AS mom_growth,
```

```
    -- Fixed: Cast the running total aggregation to numeric before rounding
    ROUND(
        (SUM(revenue) OVER (ORDER BY month))::numeric,
        2
    ) AS running_revenue
FROM monthly_sales;
```

Insight

The view successfully summarizes monthly orders, revenue, profit, month-over-month growth, and running revenue. Total running revenue reached $4,484,287.80 by Dec 2025. The strongest month was Oct 2025 with $355,440.89 revenue, while the biggest growth was Jul 2025: +108%.

Recommendation

Use staging.vw_monthly_sales_analysis as a clean reporting view for Power BI dashboards and monthly business reviews.

Business Impact

This view saves analysis time, improves reporting consistency, and helps track revenue growth trends more easily.

Product View

```
CREATE OR REPLACE VIEW staging.vw_product_analysis AS
SELECT
    p.product_id,
    p.product_name,
    p.category,
    p.sub_category,
    p.brand,
    -- Best Practice: Cast aggregate quantities to bigint for consistent data typing
    SUM(oi.quantity)::bigint AS units_sold,
    ROUND(SUM(oi.line_total)::numeric, 2) AS revenue,   -- Fixed cast
    ROUND(SUM(oi.line_profit)::numeric, 2) AS profit,   -- Fixed cast
    -- Fixed: Cast the aggregate profit margin division math to numeric before rounding
    ROUND(
        (SUM(oi.line_profit) / NULLIF(SUM(oi.line_total), 0))::numeric,
        4
    ) AS profit_margin
FROM public.order_items oi
```

```
JOIN public.products p
ON oi.product_id = p.product_id
JOIN public.orders o
ON oi.order_id = o.order_id
WHERE o.order_status <> 'Cancelled'
GROUP BY
    p.product_id,
    p.product_name,
    p.category,
    p.sub_category,
    P.brand;
```

Insight

The view staging.vw_product_analysis successfully summarizes product performance by units sold, revenue, profit, and profit margin. Nivea Fragrance Item 3 is the strongest product with $303,651.54 revenue and 45.49% margin. Some products have high revenue but lower margins, such as Bata Women Clothing Item 44 with only 17.63% margin.

Recommendation

Use this view for Power BI product dashboards. Promote high-revenue and high-margin products, and review low-margin products for pricing or discount control.

Business Impact

This view helps identify profitable products, improve pricing decisions, and support better inventory and marketing planning.

Customer RFM View

```
CREATE OR REPLACE VIEW staging.vw_rfm_analysis AS
WITH rfm_base AS (
    SELECT
        c.customer_id,
        c.customer_name,
        c.city,
        c.acquisition_channel,
```

```
        MAX(o.order_date) AS last_purchase_date,
        -- Fixed: Cast to DATE first so the subtraction yields an integer natively
        (DATE '2026-01-01' - MAX(o.order_date)::DATE) AS recency_days,
        COUNT(DISTINCT o.order_id) AS frequency,
        ROUND(SUM(o.net_revenue)::numeric, 2) AS monetary_value -- Fixed cast
    FROM public.customers c
    JOIN public.orders o
    ON c.customer_id = o.customer_id
    WHERE o.order_status <> 'Cancelled'
    GROUP BY
        c.customer_id,
        c.customer_name,
        c.city,
        c.acquisition_channel
),
rfm_score AS (
    SELECT
        *,
        -- Fixed: Flipped NTILE sorting so 5 is always the best score
        NTILE(5) OVER (ORDER BY recency_days ASC) AS recency_score,
        NTILE(5) OVER (ORDER BY frequency DESC) AS frequency_score,
        NTILE(5) OVER (ORDER BY monetary_value DESC) AS monetary_score
    FROM rfm_base
)
SELECT
    customer_id,
    customer_name,
    city,
    acquisition_channel,
    last_purchase_date,
    recency_days,
    frequency,
    monetary_value,
    recency_score,
    frequency_score,
    monetary_score,
    (recency_score + frequency_score + monetary_score) AS rfm_total_score,
    CASE
         WHEN recency_score >= 4 AND frequency_score >= 4 AND monetary_score >= 4 THEN
'Champions'
        WHEN frequency_score >= 4 AND monetary_score >= 4 THEN 'Loyal Customers'
        WHEN recency_score >= 4 AND frequency_score <= 2 THEN 'New Customers'
        WHEN recency_score <= 2 AND frequency_score >= 3 THEN 'At Risk'
        WHEN recency_score <= 2 AND frequency_score <= 2 THEN 'Lost Customers'
```

```
        ELSE 'Regular Customers'
    END AS rfm_segment
FROM rfm_score;
```

Insight

The view is created successfully, but the scoring logic is still reversed. Recent, frequent, and high-spending customers are being marked as Lost Customers, while older low-frequency customers are marked as Champions. Example: customers with very recent purchase dates still appear as Lost Customers in the output.

Final Key Insights

The business generated strong revenue and profit, with $4,484,287.80 revenue and $1,464,148.15 profit. The profit margin is healthy at 32.65%, but the 7.35% return rate needs attention.

Mobile Accessories and Audio are the strongest revenue-generating sub-categories, while Hair Care and Kitchen provide stronger profit margins. Nivea Fragrance Item 3 is the strongest product overall, but it also appears in quality risk analysis, meaning high revenue products should still be monitored for refunds and customer satisfaction.

Ramadan Offer is the only marketing campaign with strong ROAS. Mobile traffic performs best in funnel conversion, especially Google Mobile, Instagram Mobile, and YouTube Mobile. Referral App and some desktop channels have high cart abandonment.

Returns are mostly caused by damaged products, late delivery, and quality issues. Electronics has the highest unit return rate, making it a priority for return reduction.

Inventory is currently healthy, with no immediate stockout risk, but high-turnover products should be monitored for early restocking.

Customer retention needs improvement because most cohorts show low repeat purchase activity after the first purchase month.